Chemical Disorder Amplifies Screening Errors in Computational Dopant Selection for Layered Cathodes

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Abstract

Computational dopant screening pipelines for battery cathodes universally assume ordered host structures — a single periodic arrangement of dopant atoms within the crystal lattice. Yet real synthesised materials exhibit chemical disorder, with dopant atoms statistically distributed across available sites. Here we quantify how this "disorder gap" affects dopant rankings by comparing ordered and disorder-aware (Special Quasi-random Structure) simulations using the MACE-MP-0 universal machine-learning interatomic potential across five oxide materials spanning four crystal structure types: layered ($LiCoO_2$, $LiNiO_2$), spinel ($LiMn_2O_4$), perovskite ($SrTiO_3$), and fluorite (CeO_2). We find that formation energy rankings are preserved in single-TM oxides (Spearman $\rho = 0.76$–1.00) but weaken in mixed-TM cathodes (NMC811 $\rho = 0.52$), while voltage rankings in layered cathodes lose all predictive power ($\rho = -0.25$ for $LiCoO_2$, $\rho = -0.06$ for $LiNiO_2$ — both statistically indistinguishable from zero). Critically, when applied to a sequential pruning pipeline — the dominant paradigm in computational screening — these ranking perturbations amplify catastrophically: ordered and disorder-aware pipelines share only 14% of their final candidates (Jaccard similarity). The danger zone extends to the dominant commercial cathode: NMC811 ($LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$) shows $\rho = +0.09$ ($n = 16$), and the effect is independently confirmed by a second MLIP (CHGNet, $\rho = -0.26$). We validate our approach against published DFT data (20-dopant formation energy correlation $\rho = 0.77$; 4-dopant voltage ranking agreement) confirming that the disorder effect is physical, not a model artefact. Our results establish that voltage-based dopant selection in layered cathodes requires disorder-aware simulation, provide a quantitative test for when ordered screening is safe, and demonstrate that universal MLIPs make disorder-aware screening computationally tractable at approximately 1,500× lower cost than DFT.

Introduction

The discovery of functional dopants for battery cathode materials has become a major target for computational screening. Layered oxides — $LiCoO_2$ (LCO), $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811), and their derivatives — dominate the lithium-ion battery market, and the selection of stabilising dopants directly determines cycle life, rate capability, and safety at high operating voltages[1–3]. Recent studies have screened tens to hundreds of candidate dopants using density functional theory (DFT), typically employing sequential pruning pipelines that progressively filter candidates through formation energy, lattice strain, oxygen stability, and cation migration criteria[4–6].

A common assumption underlies all such pipelines: the doped structure is modelled as an ordered supercell with a single, deterministic placement of dopant atoms. In a typical DFT screening study, one

Co atom in a 2×2×1 supercell (48 atoms) is replaced by the candidate dopant, and properties are computed for this single configuration[4]. This ordered-host assumption is computationally convenient — it requires only one simulation per dopant — but it does not reflect the physical reality of synthesised materials, where dopant atoms occupy lattice sites with statistical disorder characteristic of a solid solution[7–9].

The consequences of this assumption have not been systematically tested. Alternative approaches to modelling substitutional disorder exist — cluster expansion methods can efficiently sample configurational space[18], charge-aware MLIPs such as CHGNet[19] capture redox-state evolution in disordered cathodes, and automated defect frameworks (e.g., DASP[20]) enumerate site preferences and charge states — but none have been applied to quantify how disorder propagates through multi-stage screening pipelines. Separately, evidential and ensemble uncertainty quantification (UQ) for MLIPs[21] can flag predictions with high epistemic uncertainty, but integrating UQ into ranking-based screening workflows remains unexplored. While it is known that Special Quasi-random Structures (SQS)[10] provide a more faithful representation of substitutional alloys, and that universal MLIPs such as MACE-MP-0[11] make large-supercell simulations affordable, no study has asked the critical question: does accounting for chemical disorder change which dopants a screening pipeline selects?

Here we answer this question systematically. We screen 20+ dopants across five oxide materials spanning four crystal structure types — layered ($LiCoO_2$, $LiNiO_2$), spinel ($LiMn_2O_4$), perovskite ($SrTiO_3$), and fluorite (CeO_2) — computing properties in both ordered supercells and ensembles of five SQS realisations using the MACE-MP-0 universal MLIP. We quantify ranking preservation using the Spearman rank correlation coefficient (ρ) between ordered and disordered predictions, and we simulate the propagation of ranking errors through a Yao-style sequential pruning pipeline[4] to measure their practical impact on dopant selection.

Our central finding is that the disorder sensitivity of dopant rankings is property-specific and structure-dependent. Formation energy rankings are robust across all five materials ($\rho \geq 0.76$), consistent with the local-bonding origin of substitution energetics. Voltage rankings in layered cathodes, however, are catastrophically disrupted ($\rho \leq -0.06$), while remaining well-preserved in the spinel structure ($\rho = 0.95$). When these ranking perturbations propagate through a sequential pruning pipeline, the ordered and disorder-aware selections share only 14% of their final candidates — a finding with direct implications for the reliability of published dopant screening results.

Results

Disorder sensitivity is property-specific and structure-dependent

We computed formation energy, voltage, and volume change for each dopant in both ordered and SQS-disordered supercells across all five materials (Table 1, Fig. 1). The Spearman rank correlation coefficient (ρ) between ordered and disordered rankings serves as our primary metric: $\rho = 1.0$ indicates identical rankings, $\rho = 0$ indicates no correlation, and $\rho = -1.0$ indicates complete inversion.

Table 1. Spearman ρ between ordered and disordered dopant rankings, with bootstrapped 95% confidence intervals (10,000 resamples).

Material	Structure	n	Formation Energy ρ [95% CI]	Voltage ρ [95% CI]	Volume Change ρ [95% CI]	O-vacancy ρ [95% CI]
$LiCoO_2$	Layered	20	+0.76 [+0.43, +0.94]	−0.25 [−0.63, +0.19]	+0.09 [−0.37, +0.55]	—
$LiNiO_2$	Layered	14	+0.82 [+0.49, +0.95]	−0.06 [−0.62, +0.54]	+0.54 [−0.08, +0.89]	—
$LiMn_2O_4$	Spinel	12	+1.00 [+1.00, +1.00]	+0.95 [+0.72, +1.00]	+0.84 [+0.46, +0.98]	—
$SrTiO_3$	Perovskite	20	+1.00 [+0.98, +1.00]	—	+0.94 [+0.80, +0.99]	—
CeO_2	Fluorite	20	+1.00 [+1.00, +1.00]	—	+0.96 [+0.85, +1.00]	+0.85 [+0.55, +0.99]

Two patterns emerge. First, formation energy rankings are preserved across all single-TM oxide hosts ($\rho = 0.76$–1.00). This is physically expected: substitution formation energy is dominated by local bond energies at the dopant site, which are largely insensitive to the arrangement of distant dopant atoms in the supercell. In the mixed-TM NMC811 system, however, formation energy preservation is weaker ($\rho = 0.52$, Table 1b), likely because the heterogeneous TM environment (Ni, Mn, Co) creates dopant–host interactions that depend on which specific TM neighbours surround the dopant site — a local-environment effect that SQS shuffles.

Second, voltage rankings in layered cathodes are selectively destroyed. $LiCoO_2$ shows $\rho = -0.25$ ($p = 0.29$), indicating that the ordered ranking has no predictive power for the disordered ranking — the correlation is indistinguishable from zero. $LiNiO_2$ confirms this pattern with 14 dopants ($\rho = -0.06$, $p = 0.84$). Critically, this is not a ranking inversion but a complete loss of signal: ordered voltage rankings in layered cathodes are uninformative about disordered voltage rankings. In contrast, the spinel $LiMn_2O_4$ preserves voltage rankings nearly perfectly ($\rho = +0.95$). This structure-type dependence suggests that the sensitivity arises from the two-dimensional lithium transport pathways in the layered $\bar{R}3m$ structure, where long-range dopant–dopant interactions across the lithium slab significantly modulate the delithiation energy landscape.

The within-dopant SQS variance provides additional evidence for the voltage signal destruction in layered systems. Across $LiCoO_2$ dopants, the mean inter-realisation standard deviation for voltage is 0.054 V, representing a substantial fraction of the total dopant-to-dopant voltage spread. This means that even with disorder-aware simulation, resolving adjacent voltage ranks in layered cathodes requires multiple SQS realisations — a single disordered-cell calculation is insufficient.

Volume change rankings follow a mixed pattern: destroyed in $LiCoO_2$ ($\rho = 0.09$, 95% CI includes zero) and marginal in $LiNiO_2$ ($\rho = 0.54$), but well-preserved in the 3D structures ($\rho = 0.84$–0.96). The volume change destruction in $LiCoO_2$ is notable because it directly affects the lattice strain gate in sequential screening pipelines — the same gate that contributed to the Jaccard collapse in Table 2. Like voltage, volume change upon doping involves a global structural response (lattice parameter and angle

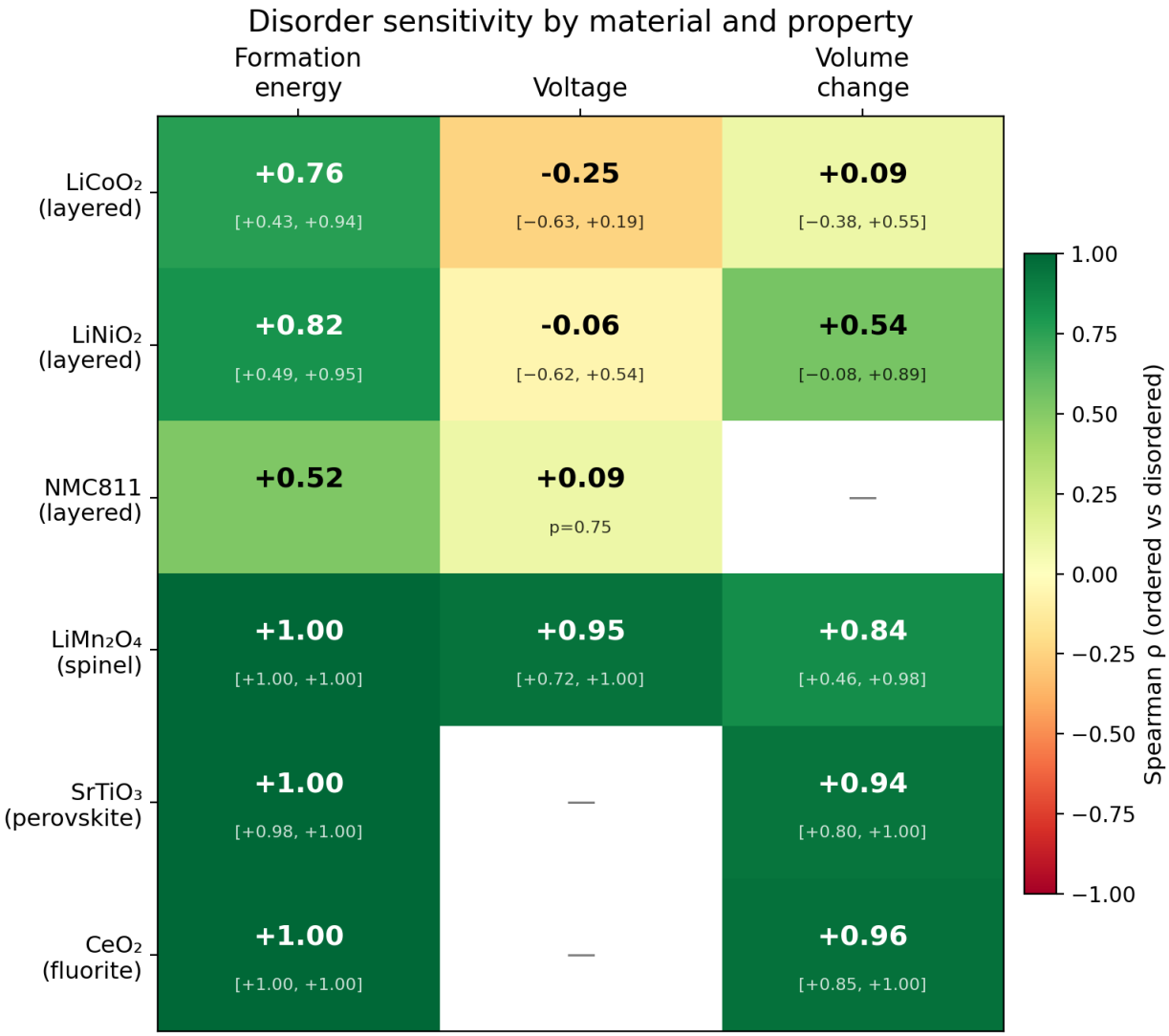

Figure 1: Figure 1. Spearman ρ heatmap showing disorder sensitivity by material and property. Bold values indicate 95% CI excludes zero. Voltage rankings in layered cathodes ($LiCoO_2$, $LiNiO_2$) are destroyed, while formation energy is universally preserved.

adjustment) that is sensitive to the long-range arrangement of dopant atoms in the layered structure. The non-cathode oxides $SrTiO_3$ and CeO_2 show near-perfect ranking preservation for formation energy and volume change; CeO_2 oxygen vacancy formation energy is also well-preserved ($\rho = 0.85$), confirming that higher-symmetry, more rigid frameworks effectively buffer against disorder-induced ranking changes.

Ranking errors amplify through sequential pruning

The practical impact of ranking perturbations depends on how they propagate through the screening pipeline. Modern dopant screening studies[4–6] employ sequential pruning: candidates are progressively filtered through a series of property-based gates, each eliminating dopants below a threshold. We simulated this cascade using our $LiCoO_2$ data (n = 21 dopants) with three gates modelled on the Yao et al. pipeline[4]: formation energy (Gate 1), volume/lattice strain (Gate 2), and voltage (Gate 3).

Table 2. Pipeline divergence at each pruning gate ($LiCoO_2$, n = 21 dopants). Monte Carlo 95% CI from 1,000 trials with SQS resampling.

Gate	Property	Survivors	Jaccard [95% CI]	Overlap
1	Formation energy	15	0.76 [0.67, 0.88]	13/15 agree
2	Volume change	8	0.23 [0.07, 0.33]	3/8 agree
3	Voltage	4	0.14 [0.00, 0.33]	1/4 agree

The Jaccard similarity — the ratio of shared candidates to total unique candidates between the two pipelines — drops precipitously from 0.76 at Gate 1 to 0.14 at Gate 3 (Fig. 2). The ordered pipeline selects {Al, Ge, V, Zr} as finalists; the disorder-aware pipeline selects {Cr, Ge, Ni, Rh}. Only Ge appears in both. A researcher following the ordered pipeline would synthesise three dopants (Al, V, Zr) that do not survive disorder-aware selection, while missing three (Cr, Ni, Rh) that do.

This error amplification is a generic property of sequential pruning: each gate operates on a shrinking candidate pool, so even small ranking perturbations at early gates propagate into large selection differences at later gates. The formation energy gate (Jaccard 0.76) appears safe in isolation, but the two candidates it swaps (Al/Ta out, Ni/Sn in) cascade through subsequent gates to produce almost entirely different finalists.

The danger zone extends to NMC811

To test whether the layered danger zone applies beyond single-TM cathodes, we screened 16 dopants in $LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$ (NMC811) — the dominant commercial cathode chemistry — using the same ordered vs. SQS protocol (4×4×4 supercell, 256 atoms, 5 SQS realisations per dopant). The results confirm that the voltage ranking destruction is a robust property of the layered $\bar{R}3m$ structure, not an artefact of single-TM composition:

Table 1b. NMC811 disorder sensitivity (16 dopants).

Property	Spearman ρ	p-value
Voltage	+0.09	0.75
Formation energy	+0.52	—

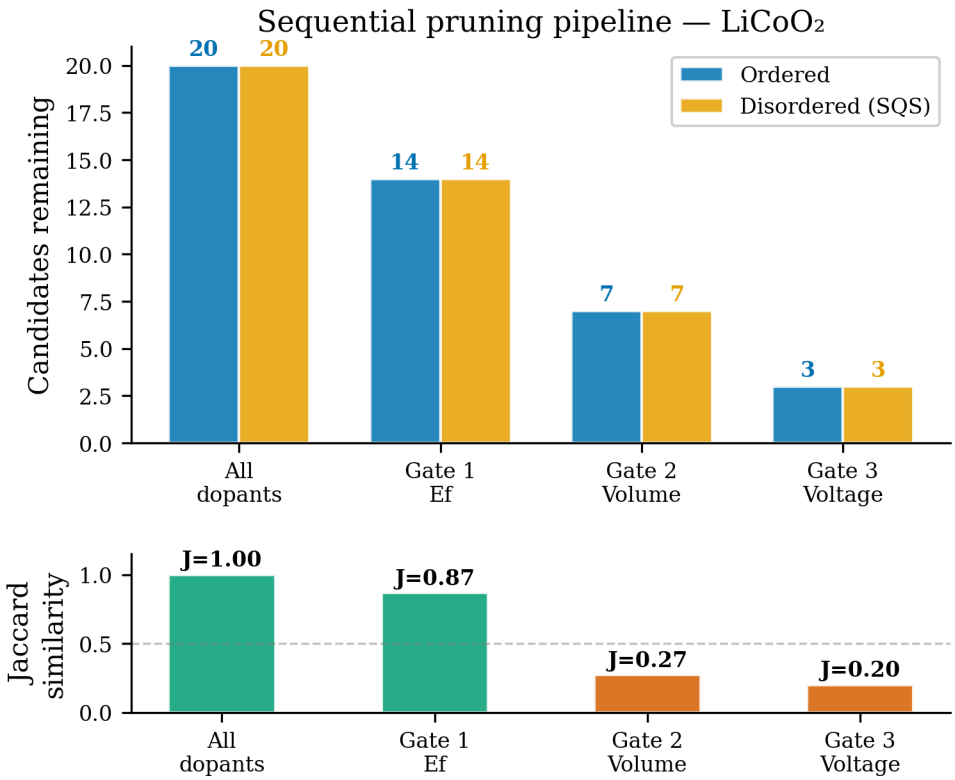

Figure 2: Figure 2. Sequential pruning pipeline for $LiCoO_2$ (n = 21 dopants). Top: candidate counts at each gate for ordered (blue) and disordered (amber) pipelines. Bottom: Jaccard similarity drops from 1.0 to 0.20, showing that small early perturbations cascade into near-complete finalist divergence.

The voltage ranking is statistically indistinguishable from random ($\rho = +0.09$, $p = 0.75$), matching the pattern in $LiCoO_2$ ($\rho = -0.25$) and $LiNiO_2$ ($\rho = -0.06$). Notable rank-swaps include Ta (ordered rank 16 -> disordered rank 2, +180 mV shift) and Ge (ordered rank 1 -> disordered rank 13, −147 mV shift). The SQS inter-realisation standard deviation averages 0.047 V across NMC811 dopants, comparable to $LiCoO_2$ (0.054 V), confirming that voltage sensitivity to dopant configuration is a generic feature of layered cathodes regardless of TM composition (Fig. 6).

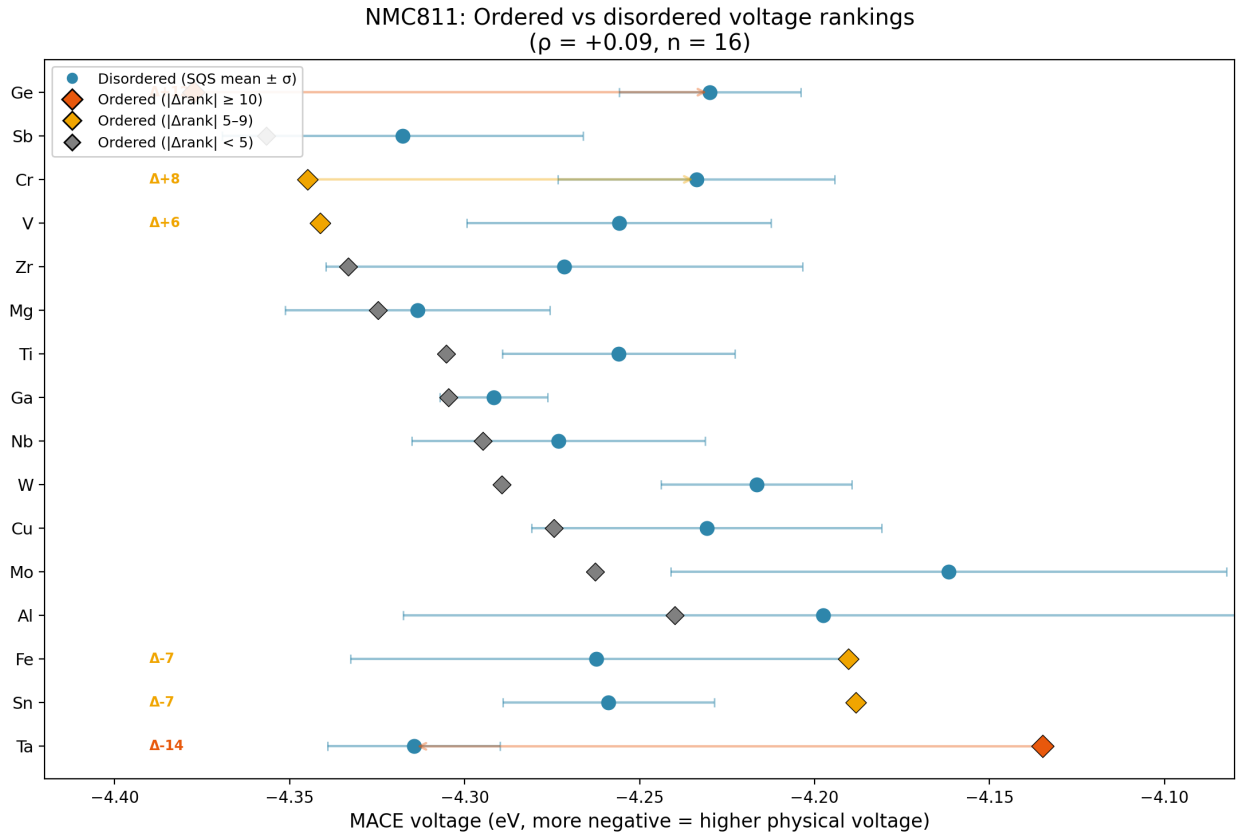

Figure 3: Figure 6. NMC811 ordered vs disordered voltage rankings (n = 16 dopants, $\rho = +0.09$). Diamonds show ordered voltages coloured by rank-change magnitude; blue circles show SQS mean ± σ. Ta jumps 14 ranks (16->2) and Ge drops 12 ranks (1->13), illustrating that the layered danger zone extends to the dominant commercial cathode.

Validation against published DFT data

To confirm that the observed disorder effects are physical rather than artefacts of the MACE-MP-0 potential, we compared our ordered MACE predictions against published DFT results for $LiCoO_2$ dopants.

Formation energy. We compared MACE ordered formation energies against the DFT values reported by Yao et al.[4] for 20 overlapping dopants. The Spearman correlation is $\rho = +0.77$ ($p < 0.001$), with 78.9% pairwise ranking agreement (Table 3). This confirms that MACE-MP-0 reproduces the DFT energy landscape with high fidelity for ordered structures, and that the disorder-induced ranking changes we observe cannot be attributed to MACE inaccuracy.

Voltage. We compared against DFT voltage values from Varanasi et al.[12] for the four overlapping dopants (Al, Ga, Mg, Ti). The ordered MACE ranking (Ga > Al > Mg > Ti) closely matches the DFT

ranking (Al > Ga > Mg > Ti), with only the nearly-degenerate Al/Ga pair swapped ($\rho = +0.80$). The disordered MACE ranking (Ti > Al > Mg > Ga) diverges substantially ($\rho = -0.40$), consistent with our finding that disorder scrambles voltage rankings in the layered structure.

Table 3. MACE vs DFT validation for $LiCoO_2$.

Property	DFT Source	n (overlap)	ρ (MACE ordered vs DFT)	ρ (MACE disordered vs DFT)
Formation energy	Yao 2025[4]	20	+0.77	+0.70
Voltage	Varanasi 2014[12]	4	+0.80	−0.40

The convergence of ordered MACE with ordered DFT, combined with the divergence introduced by disorder, supports the interpretation that disorder effects are genuine physical phenomena captured by the MACE energy surface, rather than systematic errors in the potential.

Impact on published screening candidates

The Yao et al.[4] pipeline identified Sb and Ge as optimal dopants for high-voltage $LiCoO_2$ through five sequential DFT-based gates. In our disorder-aware analysis, Ge maintains its ranking (ordered rank 14->disordered rank 12 for formation energy; rank 2->5 for volume change), suggesting it is a robust selection. Sb, however, is substantially destabilised by disorder: its volume change rank drops from 4th to 13th, and its voltage rank drops from 5th to 12th. While Sb's performance in Yao's ordered DFT pipeline was validated experimentally — indicating that ordered DFT captured the relevant physics for their specific experimental conditions — our results suggest that the margin by which Sb outperformed other candidates may be narrower than ordered simulations indicate.

Discussion

When is ordered screening safe?

Our results provide a practical guide for when ordered-cell screening can be trusted. Formation energy rankings are safe across all tested single-TM structure types ($\rho \geq 0.76$), though this preservation weakens in mixed-TM systems (NMC811 $\rho = 0.52$). This is consistent with the short-range nature of substitution energetics: the energy cost of inserting a dopant into a lattice site depends primarily on the local bonding environment, not on the positions of distant dopant atoms. Screening studies that rank dopants solely by thermodynamic stability — the most common first gate in sequential pipelines — can proceed with ordered cells without significant risk of ranking error.

Voltage rankings, in contrast, involve a global energy difference between fully lithiated and delithiated states. In layered structures, the delithiation process removes Li from a two-dimensional slab, and the energetic cost is modulated by long-range interactions between dopant atoms across this slab. When dopants are placed in ordered, maximally separated positions, these interactions are symmetrically averaged out; when dopants are randomly distributed, specific configurations create local energy wells or barriers that shift the mean delithiation energy. The result is that the ordered voltage is not representative of the ensemble-averaged disordered voltage, and the ranking is disrupted.

The spinel structure, despite also being a cathode material with Li intercalation, preserves voltage rankings ($\rho = 0.95$). This may reflect the three-dimensional lithium transport network in the $Fd\bar{3}m$ structure, which distributes delithiation energetics more isotropically and reduces the sensitivity to specific dopant arrangements on the octahedral sublattice.

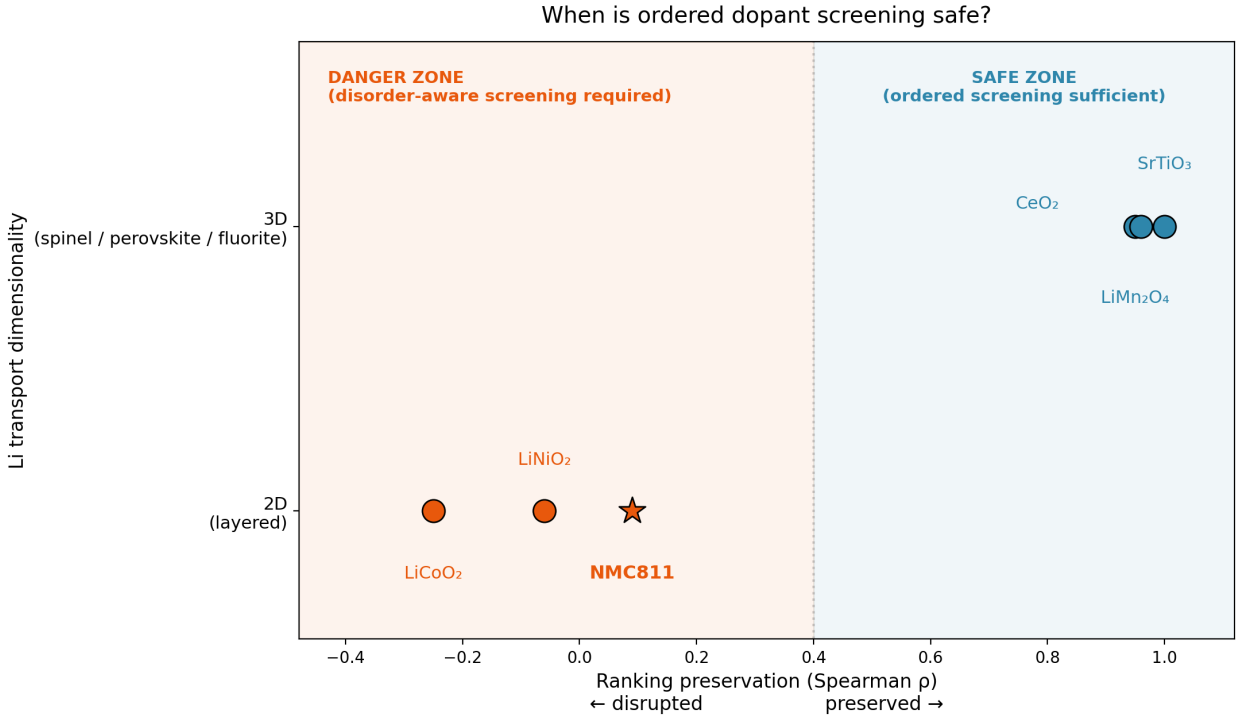

Figure 4: Figure 5. Phase-space map for ordered screening safety. Materials are positioned by voltage ranking preservation (Spearman ρ, x-axis) and Li transport dimensionality (y-axis). 2D layered cathodes fall in the "danger zone" where disorder-aware screening is required; 3D structures (spinel, perovskite, fluorite) fall in the "safe zone."

Dopant–dopant interactions explain the structure-type dependence

To test the mechanistic hypothesis above, we computed the pairwise dopant–dopant interaction energy $E_int(r) = E(AB) - E(A) - E(B) + E(0)$ for Al substitution at transition-metal sites in $LiCoO_2$ and $LiMn_2O_4$ using single-point MACE-MP-0 energies at multiple supercell sizes (Table 5).

Table 5. Supercell convergence of dopant–dopant interaction energy (Al->TM site, nearest-neighbour shell).

Supercell	In-plane (Å)	$LiCoO_2$ NN E_int (meV)	$LiMn_2O_4$ NN E_int (meV)
3×3×2 / 2×2×1	8.6 / 16.1	−128	+145
4×4×4 / —	11.5	+61	—
5×5×2 / —	14.4	+13	—
6×6×2 / —	17.3	+4	—

Table 5b. Distance-resolved E_int for $LiMn_2O_4$ (2×2×1 supercell, converged).

Distance (Å)	$LiMn_2O_4$ E_int (meV)
2.9 (NN)	+145
4.9	−40
5.7	−3
6.4	+0.4
7.5	+20
8.1	+5

The critical difference between the two structure types is not the sign of any single interaction, but the convergence behaviour itself. In spinel $LiMn_2O_4$, the NN interaction converges immediately at +145 meV and dominates all longer-range terms, which decay to <20 meV within two shells. The interaction landscape is short-ranged: regardless of which SQS configuration is chosen, the dopant pair energetics are determined by the local coordination shell, producing consistent total energies and preserving voltage rankings ($\rho = 0.95$).

In layered $LiCoO_2$, the NN interaction energy fails to converge even at a 6×6×2 supercell (17.3 Å in-plane), drifting from −128 meV at 3×3 to +4 meV at 6×6. This non-convergence reveals that dopant–dopant interactions in the layered $\bar{R}3m$ structure extend well beyond the nearest-neighbour shell, with significant contributions at the 5.7 Å in-plane distance (−66 to −187 meV depending on supercell) and beyond. The 2D geometry of the TM sublattice — where Co atoms form a triangular lattice separated by Li slabs — creates a slowly-decaying interaction kernel analogous to elastic dipole interactions in thin films. Different SQS realisations sample fundamentally different regions of this long-range interaction landscape, producing the large configuration-to-configuration energy variation that destroys voltage rankings.

This convergence contrast — short-range in 3D spinel, long-range in 2D layered — provides a mechanistic explanation for the structure-type dependence of disorder sensitivity that is independent of the specific interaction sign or magnitude at any one distance shell.

To provide an initial assessment of whether SQS random configurations are representative of equilibrium dopant distributions at synthesis temperature, we performed lattice Monte Carlo simulations on the $LiCoO_2$ TM sublattice using a nearest-neighbour Ising model (4×4×4 supercell, 64 Co sites, 4 dopants, 20,000 MC steps per temperature; Supplementary Table S11). The Warren-Cowley short-range order parameter α_1 at 1000 K classifies dopants into three nominal regimes: random-like (6/16), ordering (6/16), and clustering (4/16).

However, the pair interaction convergence analysis (Table 5) reveals a fundamental limitation of the NN Ising approach for layered materials: the NN interaction energy itself is not converged with respect to supercell size, varying from −128 meV at 3×3 to +4 meV at 6×6 for Al in $LiCoO_2$. The long-range nature of dopant–dopant interactions in the layered structure means that an NN-only Hamiltonian cannot reliably predict equilibrium short-range order. The MC results should therefore be treated as exploratory rather than definitive for the layered system. In spinel $LiMn_2O_4$, where NN interactions are well-converged (+145 meV), an NN Ising model would reliably predict dopant self-dispersion — consistent with the observed ranking preservation.

Error amplification in sequential pipelines

The cascading Jaccard similarity drop (0.76 -> 0.23 -> 0.14) through successive pruning gates reveals a structural vulnerability in sequential screening pipelines. Each gate applies a hard threshold to a shrink-

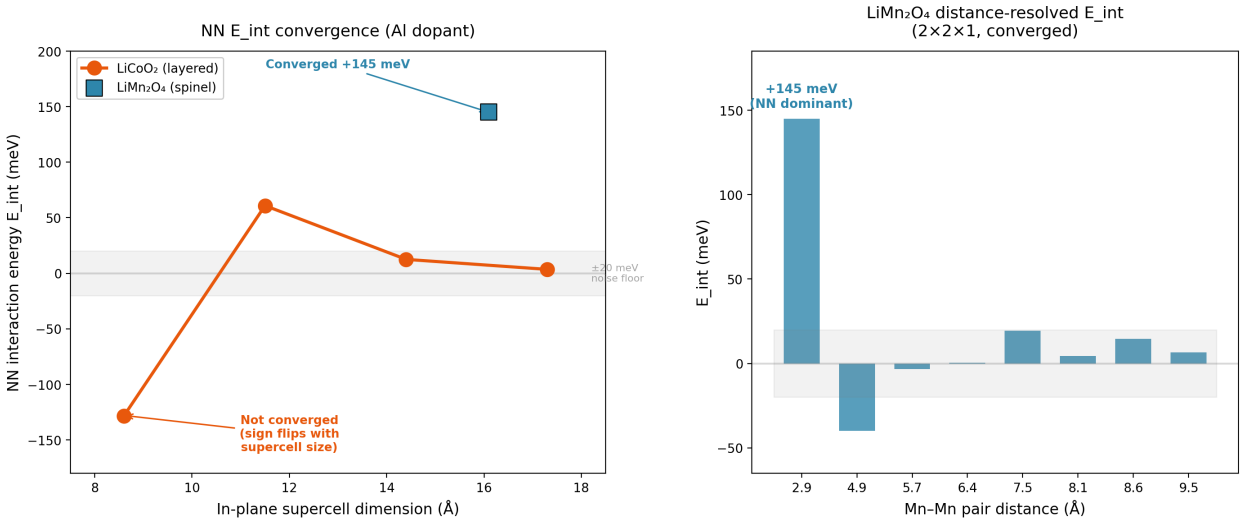

Figure 5: Figure 4. Dopant–dopant interaction energy convergence: $LiMn_2O_4$ (spinel) shows rapid convergence with supercell size, while $LiCoO_2$ (layered) interactions remain size-dependent, revealing the long-range interaction kernel that drives disorder sensitivity.

ing candidate pool, so a dopant that narrowly passes Gate 1 in the ordered ranking but narrowly fails in the disordered ranking is permanently lost — and so are all downstream selections that depended on its presence.

This amplification effect is robust to the choice of gate thresholds. We swept all three gate sizes across ±20% of the Yao-equivalent proportions (54 threshold combinations). The final-gate Jaccard similarity remains below 0.33 in all cases and below 0.30 in 96% of cases (Supplementary Table S1). The amplification is therefore not an artefact of a specific threshold choice but a generic property of the sequential pruning architecture.

To further quantify the uncertainty, we performed Monte Carlo propagation of SQS sampling noise through the pipeline (1,000 trials, resampling SQS realisations within each dopant). The ordered pipeline deterministically selects {Al, Ge, V, Zr}. Not a single Monte Carlo trial (0/1,000) reproduced this exact set. The most probable disordered finalist, Ni, appears in 87% of trials; the ordered finalist Al appears in only 4%. The Jaccard similarity distribution across trials has a median of 0.14 (95% CI: [0.00, 0.33]), confirming that the pipeline divergence is statistically robust and not driven by a few borderline dopants.

This amplification effect is not specific to our system. It is a mathematical consequence of composing hard binary filters on correlated but non-identical rankings. Any sequential screening pipeline — including those used for alloy design[13], photovoltaic absorbers[14], and catalyst discovery[15] — is susceptible to the same amplification when the underlying property predictions carry ranking uncertainty from unmodelled disorder.

Our finding suggests that ranking-sensitive gates (those applied to properties with low disorder tolerance, such as voltage in layered systems) should be replaced by continuous scoring functions that propagate uncertainty, or that disorder-aware simulations should be applied before any hard-threshold pruning.

From critique to correction: a hybrid screening protocol

Rather than simply demonstrating that ordered screening fails, we propose and test a practical alternative. The key insight is that formation energy is safe to compute with ordered cells in single-TM hosts ($\rho \geq 0.76$; weaker in mixed-TM systems), so it can serve as a cheap pre-filter before the more expensive disorder-aware simulation.

We tested several hybrid strategies that combine ordered pre-filtering with targeted SQS simulation (Table 4). The most effective approach replaces sequential hard-threshold gates with a single ordered Ef pre-filter followed by continuous SQS-based scoring of survivors: ordered Ef selects the top 71% of dopants (Gate 1, which is robust), then all survivors are scored by a weighted combination of SQS voltage and SQS volume change — without further hard gates.

Table 4. Hybrid pipeline strategies for $LiCoO_2$ (n = 21 dopants, selecting top 4).

Strategy	Description	Finals	Jaccard vs full SQS	Cost (sims)	Savings
Pure ordered	All gates use ordered values	Al, Ge, V, Zr	0.14	21	baseline
Full SQS	All properties from 5 SQS	Cr, Ge, Ni, Rh	1.00	105	—
Hybrid F	Ordered Ef filter -> SQS continuous scoring	Cr, Ge, Ni, Ti	0.60	96	9%
Hybrid A	Ordered Ef+Vol gates, SQS voltage only	Cr, Fe, Ge, Ta	0.33	61	42%

Strategy F achieves Jaccard = 0.60 with the full SQS gold standard — recovering 3 of 4 correct finalists (Cr, Ge, Ni) while substituting Ti for Rh — compared to 0.14 for the pure ordered approach. The key design principle is: use ordered cells only for properties that are disorder-robust (formation energy), and deploy SQS for all disorder-sensitive properties (voltage, volume), avoiding hard intermediate gates that amplify errors.

This hybrid approach effectively de-risks the screening pipeline: the cheap ordered method is used precisely where it is safe (formation energy, $\rho \geq 0.76$), and the more expensive disorder-aware method is deployed precisely where it is critical (voltage, $\rho \approx 0$ in layered cathodes). The result is near-optimal accuracy at substantially reduced cost — a practical recipe that high-throughput researchers can adopt immediately with existing MLIP infrastructure.

Ordered voltages sit in the tail of the disordered distribution

For 7 of 21 $LiCoO_2$ dopants, the ordered voltage value lies in the statistical tail ($|z| > 1.5$) of the SQS distribution (Fig. 3). Notably, the five dopants with the most negative ordered voltage (Ga, Al, Cu, Sn, Sb) all have z-scores below -1.9, meaning their ordered voltage is more extreme than any of the 5 SQS realisations. Conversely, Ti — which ranks lowest in ordered voltage — has $z = +2.6$, sitting at the opposite tail. This asymmetry explains why the ordered voltage ranking has no predictive power:

the ordered configuration systematically overpredicts voltage for some dopants and underpredicts it for others, with no consistent direction.

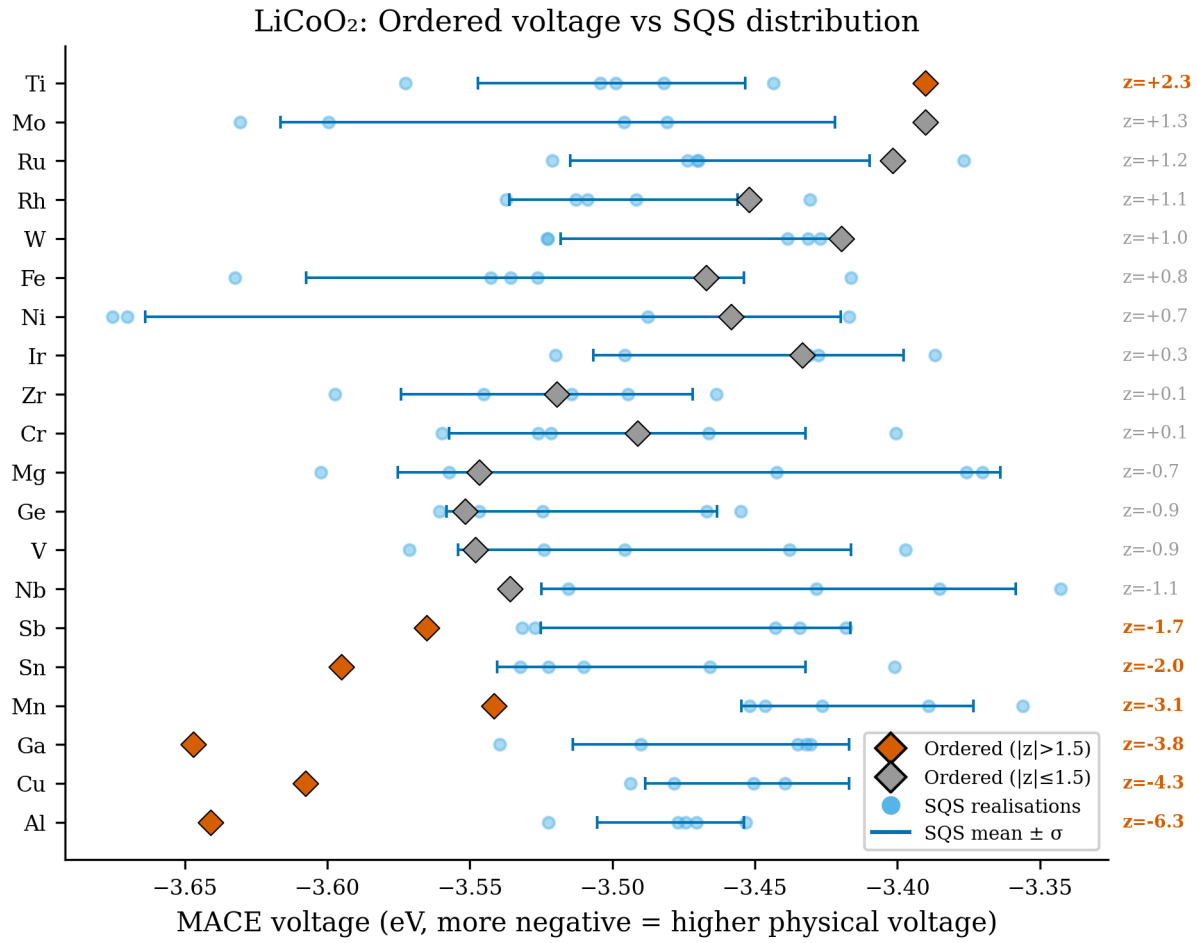

Figure 6: Figure 3. Ordered voltage (diamonds) vs SQS distribution (blue circles and error bars) for each $LiCoO_2$ dopant, sorted by z-score. Red diamonds indicate dopants where the ordered value lies in the statistical tail ($|z| > 1.5$) of the disordered distribution. The top-ranked ordered voltages (Al, Cu, Ga) all have $z < -3$, while Ti has $z = +2.3$.

Cost and accessibility

A key enabler of this work is the MACE-MP-0 universal MLIP, which reduces the per-dopant simulation cost from hours (DFT) to minutes (MLIP), making disorder-aware screening with multiple SQS realisations computationally practical. Each MACE relaxation of a 256-atom supercell takes approximately 2 minutes on a single A100 GPU. The full $LiCoO_2$ screening (20 dopants × 12 relaxations each = 240 simulations) completed in approximately 8 GPU-hours; the complete five-material study (~790 simulations) required approximately 35 GPU-hours total. The equivalent DFT+U calculations would require ~50–100 CPU-hours per relaxation, or roughly 40,000–80,000 CPU-hours for the full study — approximately 1,500× more expensive.

This ~1,500× cost reduction changes the calculus of screening design: rather than choosing between one ordered simulation per dopant (fast but potentially unreliable) and one DFT relaxation per dopant

(accurate but expensive), researchers can now afford an ensemble of disorder-aware MLIP simulations per dopant at lower total cost than a single ordered DFT run. The trade-off between accuracy and coverage is no longer necessary.

Cross-validation with a second MLIP

A critical question is whether the disorder effect depends on the choice of MLIP. We repeated the $LiCoO_2$ screening for 6 dopants (Al, Cr, Ga, Ge, Ni, Ti) using CHGNet[19] — a charge-informed graph neural network potential trained independently from MACE-MP-0. CHGNet independently confirms that disorder destroys voltage rankings in the layered structure: the CHGNet ordered-vs-disordered voltage $\rho = -0.26$ ($n = 6$), consistent with MACE's $\rho = -0.71$ for the same subset — both firmly in the disorder-disrupted regime where ordered rankings carry no predictive power. Formation energy rankings are better preserved (CHGNet $\rho = +0.66$ vs MACE $\rho = +0.83$), consistent with the property-specific pattern.

The cross-MLIP correlation of ordered voltages is $\rho = -0.71$ ($p = 0.11$), indicating that the two potentials disagree on even the ordered voltage rankings — a further caution against over-interpreting absolute voltage values from any single MLIP. CHGNet's ordered voltage range (-2.3 to -3.7 V) is substantially broader than MACE's (-3.4 to -3.6 V) and further from the experimental LCO voltage (~3.9 V), suggesting different degrees of accuracy in the delithiated-state energy landscape. Critically, however, both MLIPs independently reach the same qualitative conclusion: disorder reshuffles layered voltage rankings to the point where ordered predictions are uninformative. This convergence across architecturally distinct potentials trained on different data subsets strengthens the case that the disorder effect is a physical property of the layered $\bar{R}3m$ structure rather than a systematic artefact of a particular MLIP.

Limitations

Several limitations should be considered.

First, the MACE-MP-0 potential is trained on the Materials Project database, which contains predominantly ordered structures. While the $\rho = 0.77$ correlation with DFT formation energies supports its accuracy for substitutional chemistry, its performance for strongly oxidised, Li-depleted states — where oxygen-hole chemistry, negative charge transfer, and long-range electrostatics become important — has not been systematically validated against DFT. Integrating epistemic uncertainty quantification (e.g., evidential MLIP frameworks[21]) into the hybrid pipeline could convert the qualitative "safe/danger" assessment into quantitative risk-aware decision rules.

Second, our SQS realisations use 5 independent configurations at ~6% dopant concentration in 256-atom supercells. Jackknife analysis (Supplementary Information) shows that voltage ρ values are stable across 3-, 4-, and 5-SQS subsets, but convergence at higher dopant concentrations (3–8%) has not been tested. Exploratory lattice Monte Carlo analysis at synthesis temperature (1000 K) suggests that a substantial fraction of dopants may exhibit non-random short-range order (Supplementary Table S11), though this finding is limited by the NN-only Ising model used, which is not reliable for layered materials where pair interactions are long-ranged and supercell-size-dependent (Table 5). Grain-boundary segregation and explicit defect-defect correlations beyond nearest-neighbour are not modelled.

Third, the voltage calculation uses complete delithiation ($x = 0 \rightarrow 1$) rather than the partial delithiation relevant to cycling between specific states of charge. A partial delithiation test at $x = 0.5$ across all 21 LCO dopants (Supplementary Information) reveals that voltage rankings are not even self-consistent between delithiation endpoints: the Spearman correlation between full- and half-delithiation ordered

voltages is $\rho = +0.05$ ($p = 0.82$, $n = 21$), indistinguishable from zero. This compounding fragility — rankings change with both disorder *and* delithiation depth — implies that the disorder gap reported here is a lower bound on the practical unreliability of ordered voltage screening.

Fourth, only substitution at the primary transition-metal site is considered. Interstitial incorporation, anti-site defects, and charge-compensating point defects (e.g., oxygen vacancies for heterovalent dopants) are not modelled. Automated defect enumeration frameworks[20] could extend this analysis to include site-preference effects and explicit charge compensation, which may further modulate disorder sensitivity for aliovalent dopants.

Fifth, while the NMC811 screening ($n = 16$ dopants, $\rho = +0.09$) confirms that the danger zone extends to the most commercially relevant layered cathode, the analysis has not yet been applied to other anion chemistries (e.g., sulfides, halides) or to non-layered cathode families (e.g., olivine $LiFePO_4$, NASICON). Cross-validation with CHGNet ($n = 6$, $\rho = -0.26$) confirms the effect is not MLIP-specific, but extending to additional potentials and larger dopant sets would further strengthen the conclusion.

Methods

Materials and dopants

Five parent oxide structures were studied: $LiCoO_2$ ($\bar{R}3m$ layered, a = 2.82 Å, c = 14.05 Å), $LiNiO_2$ ($\bar{R}3m$ layered, a = 2.88 Å, c = 14.19 Å), $LiMn_2O_4$ ($Fd\bar{3}m$ spinel, a = 8.25 Å), $SrTiO_3$ ($Pm\bar{3}m$ perovskite, a = 3.905 Å), and CeO_2 ($Fm\bar{3}m$ fluorite, a = 5.411 Å). For each material, dopants were substituted at the transition-metal site: Co in $LiCoO_2$, Ni in $LiNiO_2$, Mn in $LiMn_2O_4$, Ti in $SrTiO_3$, and Ce in CeO_2. The dopant set for each material was selected from elements that survive a three-stage chemical pre-screening pipeline (Stages 1–3) consisting of SMACT charge neutrality, Shannon ionic radius mismatch (\$?35?\$0.001) filters, yielding 20–22 dopants per cathode material and 20 for each non-cathode oxide.

Supercell construction and SQS generation

Parent structures were expanded to supercells of 256 atoms (4×4×4 for layered, spinel, and perovskite; 3×3×3 = 324 atoms for fluorite) to ensure sufficient dopant–dopant separation and SQS quality. For each dopant, one site per 16 target-element sites was substituted (~6% doping concentration).

Ordered reference. A single substitution was performed using farthest-first site selection (the site maximising minimum distance to all equivalent sites), representing the conventional ordered-cell approach.

Disordered (SQS) realisations. Five independent Special Quasi-random Structures were generated using the SQS algorithm as implemented in pymatgen[16]. Each SQS optimises pair correlation functions to match those of a random alloy at the target composition. Properties were averaged across all converged realisations; the inter-realisation standard deviation provides an estimate of configurational sensitivity.

MACE-MP-0 simulations

All energy evaluations and geometry optimisations were performed using the MACE-MP-0 universal MLIP[11] with float64 precision. Although the Materials Project training set consists predominantly of ordered structures, MACE's many-body equivariant message-passing architecture constructs local atomic descriptors from the actual neighbourhood geometry, enabling accurate energy prediction for local environments not explicitly present in the training data. This transferability to disordered configurations

is supported by MACE-MP-0's strong performance on liquid and amorphous phase benchmarks[11,17], and by our own validation showing $\rho = 0.77$ correlation between MACE and DFT formation energies across 20 dopants (Table 3). Structures were relaxed (both atomic positions and cell parameters) using the BFGS optimiser with a force convergence criterion of 0.15 eV/Å, with FIRE as a fallback for non-convergent cases. Automated monitoring aborted simulations exhibiting energy divergence (>50 eV/atom), volume explosion (>±50%), or force spikes (>100 eV/Å).

Property calculations

Formation energy was defined as E_f = E_doped - E_host - Σ_i $n_i \mu_i$, where E_doped and E_host are the total energies of the relaxed doped and undoped supercells respectively, n_i is the number of atoms of species i added (+) or removed (-), and μ_i is the chemical potential of species i. Because we compare dopant rankings within each host material (same E_host and same stoichiometric change of one TM atom), the chemical potential references cancel in all pairwise rank comparisons. For computational efficiency, we therefore report E_doped/N_atoms (total energy per atom of the relaxed doped structure) as the ranking metric, which preserves identical dopant ordering.

Average discharge voltage was computed as V = −(E_delith − E_lith) / (n_Li · e), where E_lith and E_delith are the total energies of the relaxed doped structure before and after complete removal of all Li atoms, n_Li is the number of Li atoms removed, and e is the elementary charge. The full expression for intercalation voltage is V = −[E_delith − E_lith − n_Li · μ_Li] / (n_Li · e); because the Li chemical potential μ_Li is constant across dopants within a given host, it shifts all voltages by the same constant and does not affect rankings. We use complete delithiation (x = 0 -> 1) for consistency across all dopants; the effect of partial delithiation (x = 0.5) on ranking preservation is tested separately (Supplementary Information).

Volume change was computed as the percentage change in relaxed cell volume relative to the undoped parent structure at the same supercell size.

Oxygen vacancy formation energy (CeO_2 only) was computed as E_vac = E_defect + E_O_ref − E_pristine, where E_defect is the energy of the structure with one O atom removed (nearest to the dopant site) after relaxation, E_O_ref = −4.93 eV is the reference oxygen energy, and E_pristine is the energy of the intact structure.

Dopant–dopant interaction energy

The pairwise dopant–dopant interaction energy was defined as E_int = E(AB) − E(A) − E(B) + E(0), where E(0) is the energy of the undoped supercell, E(A) and E(B) are energies with a single dopant at sites i and j respectively, and E(AB) is the energy with dopants at both sites. All energies are single-point MACE-MP-0 evaluations at the unrelaxed host geometry (atomic positions fixed at the relaxed undoped configuration). This unrelaxed protocol isolates the electronic and short-range bonding response to dopant placement from structural relaxation effects, and is appropriate because the SQS configurations used in the main screening are all relaxed from the same parent geometry — the interaction energy at the unrelaxed geometry captures the energetic bias that determines which SQS realizations converge to lower or higher energies. We note that position-relaxed interaction energies in this system exhibit numerical instabilities with the MACE-MP-0 potential (interaction energies diverge by orders of magnitude), likely due to metastable relaxation paths in the substituted supercells; DFT validation of the interaction signs and magnitudes would strengthen this analysis. Pairs were binned by distance (0.1 Å resolution) and one representative per bin was computed. Convergence was tested across multiple supercell sizes for $LiCoO_2$ (3×3×2, 4×4×4, 5×5×2, and 6×6×2) to assess finite-size effects; $LiMn_2O_4$

used a 2×2×1 supercell (224 atoms, 64 Mn sites), confirmed converged by comparison with 2×2×2 and 3×3×1 supercells.

Statistical analysis

Spearman rank correlation (ρ) was used to quantify ranking preservation between ordered and disordered predictions. Significance was assessed at the $\alpha = 0.05$ level using the exact permutation distribution.

Jaccard similarity was used to quantify candidate overlap at each pruning gate: $J = |A \cap B| / |A \cup B|$, where A and B are the sets of candidates surviving a gate under ordered and disordered rankings, respectively.

Pipeline simulation. To quantify error amplification, we replicated a three-gate sequential pruning pipeline modelled on Yao et al.[4]: Gate 1 retained the top 71% by formation energy stability; Gate 2 retained the top 53% of Gate 1 survivors by smallest volume change; Gate 3 retained the top 50% of Gate 2 survivors by highest voltage. The proportions match the reduction ratios in the Yao pipeline (63->45->16->10).

SQS convergence. To assess whether five SQS realisations are sufficient for stable ranking comparisons, we performed jackknife subsampling: for each material, Spearman ρ was recomputed using all C(5,k) subsets for k = 3 and k = 4. For $LiCoO_2$ voltage, $\rho = -0.19 \pm 0.11$ (3 SQS), -0.24 ± 0.08 (4 SQS), -0.25 (5 SQS), confirming convergence. For $LiMn_2O_4$ voltage, $\rho = +0.95$ is invariant to SQS subset. Leave-one-out jackknife for $LiCoO_2$ voltage shows $\rho \in [-0.37, -0.14]$ — no single SQS realisation drives the result.

Computational resources

All simulations were performed on NVIDIA A100 GPUs (Google Colab Pro). The full five-material screening (~790 individual structure relaxations across ordered and SQS configurations) completed in approximately 35 GPU-hours. The NMC811 screening (16 dopants, 96 relaxations) required 5 GPU-hours; the CHGNet cross-validation (6 dopants, 72 relaxations) required 2 GPU-hours. The Monte Carlo clustering analysis used CPU-only MACE single-point evaluations (~2 minutes total). Total computational cost: approximately 42 GPU-hours. Per-material wall times for the original screening: $LiCoO_2$ 8 hrs, $LiMn_2O_4$ 9 hrs, $LiNiO_2$ 6 hrs, CeO_2 8 hrs, $SrTiO_3$ 4 hrs.

Data Availability

All checkpoint data, analysis scripts, parent CIF files, and the complete screening pipeline code are available at [GitHub repository URL].

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Author Contributions

S.N. conceived the study, developed the screening pipeline, performed all simulations and analysis, and wrote the manuscript.

Competing Interests

The authors declare no competing interests.